

KUDYMKAR, Russian Federation

WMO: 281160

Lat: 59.017N

Lon: 54.650E

Elev: 610

StdP: 14.37

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.7	-19.5	-30.8	1.0	-25.4	-24.4	1.5	-19.3	16.8	12.9	14.9	12.4	3.5	320	0.619

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	83.6	68.3	80.2	66.6	76.8	64.7	70.2	80.3	68.4	77.5	66.5	74.4	5.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.7	100.6	75.6	65.0	94.6	73.2	63.3	89.0	71.0	34.5	80.2	33.0	77.6	31.4	74.6	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	14.4	12.8	-32.5	88.2	7.3	2.9	-37.7	90.3	-42.0	92.0	-46.0	93.6	-51.3	95.8
			-32.7	72.8	6.8	2.9	-37.6	75.0	-41.6	76.7	-45.4	78.3	-50.3	80.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.7	7.0	9.6	23.4	37.4	50.0	59.2	63.6	58.8	48.6	36.4	21.2	11.5	(6)
(7)		DBStd	22.30	14.12	13.55	8.82	9.17	9.77	7.77	7.63	7.68	8.02	8.36	11.94	14.12	(7)
(8)		HDD50	6434	1333	1131	824	388	128	13	3	10	120	429	863	1192	(8)
(9)		HDD65	10838	1798	1551	1289	827	472	205	121	224	494	887	1313	1657	(9)
(10)		CDD50	1221	0	0	0	11	127	291	424	283	78	7	0	0	(10)
(11)		CDD65	151	0	0	0	0	7	33	77	32	2	0	0	0	(11)
(12)		CDH74	1368	0	0	0	1	106	278	703	268	12	0	0	0	(12)
(13)		CDH80	317	0	0	0	0	16	47	177	75	2	0	0	0	(13)
(14)	Wind	WSAvg	5.7	5.4	5.6	6.3	6.1	6.7	5.6	4.9	5.0	5.5	6.2	5.7	5.5	(14)
(15)	Precipitation	PrecAvg	23.5	1.6	1.0	1.4	1.3	1.8	3.0	2.9	2.6	2.2	2.1	2.0	1.7	(15)
(16)		PrecMax	30.0	2.6	2.3	3.0	2.5	4.0	5.7	6.3	5.9	5.0	5.3	4.3	2.7	(16)
(17)		PrecMin	18.6	0.5	0.2	0.2	0.0	0.8	0.4	0.3	0.7	0.4	0.6	0.4	0.8	(17)
(18)		PrecStd	3.0	0.6	0.5	0.7	0.7	0.8	1.4	1.6	1.2	1.1	1.0	0.8	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.0	34.2	47.1	69.5	82.1	84.7	87.6	87.4	74.9	62.9	47.0	36.3	(19)
(20)		0.4%	MCWB	31.9	32.7	39.1	53.2	61.8	68.9	70.2	70.1	60.4	52.0	44.9	35.6	(20)
(21)		2%	DB	30.9	32.7	40.7	61.5	76.9	80.5	84.6	80.7	68.8	56.3	40.8	33.7	(21)
(22)		2%	MCWB	30.0	31.6	34.3	50.6	59.2	66.7	69.1	67.7	58.9	49.4	39.4	32.7	(22)
(23)		5%	DB	28.6	30.7	37.2	55.8	72.1	77.1	81.5	75.8	64.1	51.6	36.6	31.8	(23)
(24)		5%	MCWB	27.8	29.6	32.9	46.1	56.9	64.9	68.2	65.2	57.1	47.4	35.5	31.1	(24)
(25)		10%	DB	25.6	27.9	34.7	51.1	67.5	73.3	77.9	71.7	60.6	48.1	34.0	29.0	(25)
(26)		10%	MCWB	24.7	26.8	31.9	43.0	54.8	62.8	66.5	63.2	55.7	45.7	33.0	28.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.4	33.5	40.0	55.0	64.8	71.1	73.2	72.2	62.8	53.9	45.1	35.9	(27)
(28)		0.4%	MCDWB	32.9	33.9	46.6	65.4	77.5	80.6	84.2	85.3	69.6	59.5	46.5	36.1	(28)
(29)		2%	WB	30.1	31.8	35.7	51.0	61.3	68.8	71.0	68.7	60.5	50.4	39.5	33.0	(29)
(30)		2%	MCDWB	30.9	32.4	38.5	60.9	73.3	78.0	81.2	77.9	66.3	55.0	40.5	33.6	(30)
(31)		5%	WB	27.9	29.6	34.0	47.1	58.5	66.5	69.2	66.4	58.1	47.7	35.7	31.2	(31)
(32)		5%	MCDWB	28.6	30.7	36.2	54.5	68.7	74.6	78.8	73.7	63.1	50.6	36.5	31.9	(32)
(33)		10%	WB	24.8	26.9	32.3	43.9	55.9	64.0	67.5	64.1	55.8	45.6	33.3	28.1	(33)
(34)		10%	MCDWB	25.5	28.0	34.4	50.1	65.8	71.4	76.0	70.7	60.1	48.2	33.9	28.9	(34)
(35)	Mean Daily Temperature Range		MDBR	11.2	13.2	13.8	16.0	19.6	17.4	18.3	15.9	13.3	8.8	8.0	9.8	(35)
(36)		5% DB	MCDBR	9.3	9.2	15.3	23.1	26.6	22.2	23.4	21.8	18.6	15.2	8.0	6.8	(36)
(37)			MCWBR	9.5	8.7	10.8	14.1	13.1	11.0	10.2	10.1	10.5	9.5	7.6	6.5	(37)
(38)			MCDBR	10.0	9.1	12.1	21.8	23.8	19.7	21.0	20.2	16.6	13.8	7.7	6.4	(38)
(39)		5% WB	MCWBR	10.2	8.7	9.4	13.9	13.3	10.8	10.5	10.1	10.2	9.8	7.6	6.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.278	0.309	0.358	0.385	0.379	0.373	0.399	0.403	0.363	0.334	0.293	0.265	(40)	
(41)		taud	2.004	2.026	1.969	2.104	2.337	2.397	2.397	2.332	2.301	2.381	2.305	2.107	1.964	(41)
(42)		Ebn at Noon	154	213	237	257	270	273	262	250	242	210	150	114	(42)	
(43)		Edh at Noon	17	28	40	42	36	34	36	34	27	21	15	12	(43)	
(44)	All-Sky Solar Radiation	RadAvg	141	403	799	1211	1569	1699	1707	1259	725	314	121	75	(44)	
(45)		RadStd	19	79	103	163	178	155	165	120	84	34	21	13	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(46)
(47)	Regional (8 neighbors)	(m) N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page